Quantitative Day Trading Strategy: Research & Specification Report

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Quantitative Research — Internal Use

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Executive Summary

This report documents a from-first-principles selection of an intraday equity day-trading strategy for deployment on the llamaOS paper-trading stack (US equities via Alpaca; no futures, no options, no leverage beyond account margin settings).

What was surveyed. Five strategy families were assessed against a verified literature dossier of seven named works: (A) Opening Range Breakout (ORB); (B) intraday momentum including gamma-hedging demand; (C) VWAP mean reversion versus trend; (D) generic OHLCV index/futures signals, including the 2026 MNQ falsification study; and (E) factor-zoo machine-learning intraday predictability. Each family was graded on five dimensions: statistical significance, transaction-cost survivability, regime dependency, sample size/robustness, and long/short symmetry.

What was selected. A long-biased, news-selected 5-minute Opening Range Breakout on “Stocks in Play” — the operational family of Zarattini, Barbon & Aziz (2024). It is the lowest-infrastructure, microstructure-supported choice that fits the existing stack: a deterministic 1-minute rule engine, broker-side brackets, an EOD flatten, and a news/universe pipeline already present in llamaOS. The machine-learning factor-zoo approach (Family E) scores highest on raw academic evidence but requires a 218-factor data and modeling pipeline that does not exist here; it is deferred to a separate project.

Honest evidence verdict. Our own rerunnable backtest of the *mechanical* ORB rule on a fixed 35-name basket (1,046 trades, 2024-01-01 to 2026-10-16, 10 bps round-turn costs) nets only **+$590** on a $100,000 account, a net annualized Sharpe of **0.007**, with a block-bootstrap 95% Sharpe confidence interval of **[-1.34, +1.10]** that straddles zero. On our own evidence, the fixed-basket mechanical ORB **does not reliably overcome costs**. The deployment thesis rests on the *attention-selected universe* — names with fresh company news and relative volume (RVOL ≥ 2.0) — which the standalone script did not simulate. That selectivity is the entire economic bet and must be validated in the dashboard Backtesting workspace before any paper trading.

Deployment stance: paper-first. The system ships the long side only (the short mirror showed negative expectancy on our own run). Default mode is record/paper; real-money endpoints are refused by the stack. No guaranteed-return language is used; all performance figures are labeled by provenance — authors' own backtest, our own backtest, or literature expectation.

1. Literature Survey

We separate three kinds of number throughout: **(a) authors' own backtest numbers** (as reported by the papers, not independently replicated), **(b) our own backtest numbers** (Section 4), and **(c) literature expectations** used only to set realistic ranges. Every traceable claim links to the verified dossier.

1.1 Citation verification ledger

The prompt supplied seven named works. Each was checked against an authoritative source before being used as evidence.

#	Work as named	Verdict	What it actually is
1	arXiv 2605.04004, MNQ OHLCV signals	Verified (minor title correction)	Mesfin, "Structural Limits of OHLCV-Based Intraday Momentum Signals in MNQ Futures." Working paper, not peer-reviewed.
2	"Vortex Capital" 2026 ORB asymmetry	Blog post, not a study	Vortex blog, 13 May 2026. Last 20 days on 9 high-beta names. A desk observation.
3	"Vortex" Off-Radar VWAP paper	Blog post, not a paper	Vortex, 8 Jul 2026. ~85 sessions, measured NBBO. Direction is a falsification of VWAP fades.
4	"Boller et al., Duke 2025"	Verified; name correction	Real authors: **Aleti, Bollerslev & Siggaard (2025), Management Science**. No author "Boller" exists.
5	Notre Dame gamma-hedging momentum	Verified	Baltussen, Da, Lammers & Martens (2021), JFE 142. Zhi Da at Notre Dame. Peer-reviewed.
6	Handelshögskolan (SSE) ORB thesis	Partially verified; institution misattributed	Real paper: **Holmberg, Lönnbark & Lundström (2013), Finance Research Letters** — authors at **Umeå University**, not SSE; asset is US crude oil futures, not equities.
7	"SPY intraday momentum" RePEc 2024	Verified	Zarattini, Barbon & Aziz, SFI RP 24-97; the ORB paper analyzed is SSRN 4729284 / SFI RP 24-98.

Two further named items were searched and **not found** under the stated identity: "Boller" as a single Duke author (the real author is Bollerslev), and any SSE/Handelshögskolan ORB thesis (the closest peer-reviewed paper is Umeå, not SSE). These corrections are stated plainly so this report is never cited under the wrong institution or name.

1.2 Family A — Opening Range Breakout (ORB)

- **(1) Significance:** Mixed. Zarattini/Barbon/Aziz (working paper): authors' own top-20 Sharpe **2.81**; cost-honest QQQ alpha **33% p.a. (p = 0.0025)**. Holmberg et al. (peer-reviewed, oil futures 1983-2011): bootstrap-significant gross returns. Mesfin (2026): MNQ ORB long OOS **t = 0.88** — fails.
- **(2) Cost survivability:** Weak. Zarattini uses commission only, no slippage/borrow; the selected names are exactly where slippage and hard-to-borrow fees are worst. The cost-honest number is QQQ Sharpe **1.12**, not 2.81. Holmberg assumes zero costs.
- **(3) Regime dependency:** The high-vol 2020 and meme 2021 years likely carry much of the authors' edge; the Vortex blog reports a recent long-ORB hit rate of only ~30.6%.
- **(4) Sample / robustness:** Zarattini 8 years, 1,795 QQQ trades; Holmberg 28 years of oil futures. **No independent peer-reviewed replication of the 2.81 Sharpe exists.**
- **(5) Long/short symmetry:** QQQ variant ran 51% long / 49% short. Our own run (Sec. 4) drives the deploy-long-only decision.

Key numbers: QQQ ORB Sharpe **1.12**, win rate 24%, 0.13R/trade; top-20 stocks-in-play Sharpe **2.81** (authors' own, cost-light); MNQ ORB long OOS t = 0.88 (falsified).

1.3 Family B — Intraday momentum (incl. gamma hedging)

- **(1) Significance:** Strongest family. Gao, Han, Li & Zhou (2018, JFE): first half-hour predicts the last half-hour, **t = 4.08**, in-sample R^2 = 1.6%, OOS R^2 = 1.4%. Baltussen et al. (2021, JFE): rest-of-day predicts last 30 min across 60 futures, **t = 6.46**. Heston, Korajczyk & Sadka (2010, JF): time-of-day continuation persists ~40 days.
- **(2) Cost survivability:** Good. The Gao timing trade occurs once per day; a ~1-2 bps SPY round trip is small against ~6.67% annualized gross.
- **(3) Regime dependency: Strong.** High-vol tercile R^2 = 3.3% vs low 0.6%; recessions R^2 = 6.6% vs expansions 1.0%; crisis joint R^2 = 6.9%.
- **(4) Sample / robustness:** SPY 1993-2013 (21 years) and 60 futures over 1974-2020 (~46 years).
- **(5) Long/short symmetry:** Symmetric by construction; continuation occurs on up and down days.

Key numbers: t = 4.08 (Gao), t = 6.46 (Baltussen); timing Sharpe **1.08** vs buy-and-hold 0.29; recession R^2 6.6% vs expansion 1.0%.

1.4 Family C — VWAP reversion versus trend

- **(1) Significance:** No peer-reviewed standalone VWAP-fade edge. The literature supports short-term reversal at the open, not a “fade 2-sigma from VWAP” rule.
- **(2) Cost survivability:** Actively falsified. Vortex: naive reversion 72.6% of days, but the tradable rate (reach VWAP before a 1-sigma stop) is only **35.2%**; net of measured half-spread, **-24 bps/trade**.
- **(3) Regime dependency:** Perverse — the more stretched the price, the lower the tradable hit rate (39% to 24%).
- **(4) Sample / robustness:** 29 mid-caps + 8 mega-caps, ~85 sessions, n = 833 fades; no academic replication.
- **(5) Long/short symmetry:** Symmetric; both sides fail.

This family is **not traded**.

1.5 Family D — OHLCV index/futures signals (MNQ falsification)

- **(1) Significance:** Mesfin (2026): **14 OHLCV families on MNQ, none passed all five walk-forward criteria**. ORB long t = 0.88, VVG reversal t = 1.26, gap-continuation short t = 1.46. Two positive controls (t = 3.11 and 4.30) confirm the framework detects real edge.
- **(2) Cost survivability:** 11 of 14 families produce gross 0.07-1.50 MNQ points against a **2.0-point friction floor** (~$100/contract). The friction floor binds.
- **(3) Regime dependency:** Near-misses fail on year-to-year direction inconsistency across 2023/2024/2025.
- **(4) Sample / robustness:** 947 regular-hours days, 5-min bars, 2021-2025; single asset, single author, not peer-reviewed, but a strict walk-forward protocol.
- **(5) Long/short symmetry:** Gap-continuation short nearly passes while ORB long fails, but the short sample is tiny (N = 22).

Critical caveat: MNQ futures carry a futures friction floor; this says generic OHLCV signals on index futures do not survive realistic futures costs, *not* that the same signals fail on selective US equities.

1.6 Family E — Factor-based intraday predictability

- **(1) Significance:** Aleti, Bollerslev & Siggaard (2025, Management Science): OOS 15-min R^2 = **0.212%** (vs Ross bound 0.00777%); Diebold-Mariano t = -4.315.
- **(2) Cost survivability:** Measured spreads 0.67-1.67 bps; after-cost SPY Sharpe **1.37**, FF6 alpha 20.83% p.a.
- **(3) Regime dependency:** Most alpha accrues in high-uncertainty periods (tail-risk and liquidity factors).
- **(4) Sample / robustness:** 2004-2020, 15-min bars, 218 factors, multiple models; ensemble beats individuals.
- **(5) Long/short symmetry:** Time-series SPY timing (long/flat), not cross-sectional L-S.

Key numbers: OOS R^2 0.212%; after-cost Sharpe **1.37**; FF6 alpha 20.83% p.a. Ranks first on raw evidence but is **not deployable here** — it needs a 218-factor pipeline and 15-minute rebalancing that llamaOS does not build.

1.7 Ranked evidence summary

Rank	Family	Position
1	Family E — factor-zoo ML timing	Strongest evidence; infrastructure not built
2	Family B — first/last-half-hour momentum	Lower-infra fallback; edge in high-vol regimes
3	Family A — selective ORB on stocks-in-play	**Selected**; cost-honest QQQ Sharpe 1.12
4	Family D — generic OHLCV index/futures	Falsified on futures
5	Family C — VWAP fade	No edge; actively falsified

2. Strategy Selection & Justification

2.1 The selected strategy

We deploy **a long-biased, news-selected 5-minute Opening Range Breakout on "Stocks in Play"**, following Zarattini, Barbon & Aziz (2024, SSRN 4729284). Deployable files: *strategies/orb_stocks_in_play_pro_long.v1.yaml* (long) and its gated mirror *orb_stocks_in_play_pro_short.v1.yaml* (held off).

2.2 Economic and microstructure mechanism

The thesis is **attention-driven volume leading to short-horizon momentum continuation**. A stock with fresh company news draws concentrated attention at the open; the resulting one-directional order flow persists for the first hour. **Gao et al. (2018)** show opening price discovery predicts the close (t = 4.08); **Baltussen et al. (2021)** attribute intraday momentum to options-market-maker gamma hedging and leveraged-ETF rebalancing; **Bogousslavsky (2021)** anchors it structurally to slow-moving, infrequently-rebalancing capital — which is why the effect persists across decades. The operational bet is therefore not "ORB works on any stock" — our basket test disproves that — but "ORB works on the attention-selected subset." The news pipeline and RVOL ≥ 2.0 gate are two attempts to select it.

2.3 Direct response to Mesfin (arXiv 2605.04004)

The strongest objection to deploying any OHLCV breakout is Mesfin's falsification. We respond point by point:

- **Cross-sectional selectivity vs single-instrument index OHLCV.** Mesfin tests 14 families on **MNQ**, one index future, with no universe selection. We take a **cross-sectional attention-selected** equity slice (fresh news, RVOL ≥ 2.0) — the same idea as the paper's top-20 relative-volume ranking. Selection, not the breakout shape, is the difference.
- **Equity spread vs futures friction floor.** Mesfin's binding constraint is a 2.0-point (~$100/contract) futures floor. On liquid Alpaca equities commission is **$0** and our modeled round-turn is **10 bps**, a structurally smaller drag.
- **Why this may not be enough — honestly.** Our fixed-basket test nets Sharpe 0.007 with a bootstrap CI over zero; at 2x/3x costs it loses $155k/$311k. The equity-spread advantage does not by itself manufacture an edge. The news-universe selectivity is unproven in our standalone script. We are deploying a *hypothesis*, not a demonstrated edge.

2.4 Why not the other families

- **Why not Family E (ML factor zoo)?** Best peer-reviewed after-cost Sharpe (1.37), but it needs a 218-factor pipeline and 15-minute rebalancing that llamaOS does not build. It is a separate project (Sec. 7), not a v1 deploy.
- **Why not Family C (VWAP fade)?** Actively falsified: net -24 bps/trade off-radar once spreads are charged.
- **Why not Family B as primary?** It is the fallback; its edge is explicitly high-vol concentrated, and our own data unexpectedly shows the mechanical ORB edge in the *low*-SPY-vol bucket — a tension we flag rather than resolve.

2.5 Long/short asymmetry: deploy long only

Our run split 524 long / 522 short. The long side netted **+$22,047 (+2.8 bps/trade)**; the short side netted **-$21,457 (-2.8 bps/trade)**. Shorting adds borrow cost, recall risk, and equity drift with no compensation in our sample. We

deploy **long only**; the short YAML remains in the repo but is rejected by the factory risk controls while allow_short is off — the intended default.

3. Fully Specified Trading System

All times are US Eastern (ET). Bold values are the v1 defaults in the deployable YAML.

3.1 Instrument universe

- **Universe:** news-selected US equities (*universe.from.news_today: true*) — only names with fresh company news since the prior close. The standalone backtest used a fixed 35-name basket and did **not** exercise this selection.
- **Liquidity filters (prior close):** price ≥ **$5**; 14-day avg daily dollar volume ≥ **$20,000,000**; 14-day daily ATR ≥ **$0.50**.
- **Reference instrument: SPY** (*references: [SPY]*), used only as a regime gauge and never traded. Why SPY: the most liquid broad-market proxy for risk-on/risk-off and realized-volatility percentile.

3.2 Entry rules

- **Opening range:** first **5 minutes** (09:30-09:35 ET), *or_minutes = 5*.
- **Candle direction:** the first 5-min candle must close **above the day open** for a long (short mirror requires close below open).
- **Crossing rule:** a 1-min bar must **close beyond the 5-min range high**. The backtest fills at the **next bar's open** to avoid look-ahead.
- **Relative volume: RVOL(14) ≥ 2.0** (cumulative volume vs the prior-14 mean at the same minute-of-day).
- **Entry window: 09:35-10:30 ET** only; entries after 10:30 are cut off.
- **Two default-off experimental filters:** *volrank_min = 0.0* (SPY regime gate OFF) and *candle_atr_min = 0.0* (first-candle magnitude filter OFF). Raise as experiments, not priors.

3.3 Exit rules

- **Stop: 0.10 x the 14-day daily ATR** (*stop_atr = 0.10*), the paper's own stop. A breakout that reverses by ~10% of a daily ATR has failed its premise.
- **Target: 10R** placeholder — almost never triggers; it exists only because the broker bracket needs a profit leg.
- **Time exit:** flatten all at **15:50 ET**, 5 minutes before the 15:55 risk flatten, to avoid the closing auction and end flat.
- **Trailing stop: none in v1**. Proposed v2: after +1R move stop to breakeven; after +2R trail by 0.5 x ATR14 (research suggestion only).

3.4 Sizing and risk

- **Risk per trade: 0.5%** of equity ($500 on $100k); shares = $500 / (0.10 x ATR14).
- **Frequency cap: max 3 trades/day**; one position per symbol per day.
- **Daily loss:** the account Risk page enforces *max_daily_loss = 2%*, on breach the kill switch fires. **Stated gap:** the standalone script does not simulate a per-strategy intraday daily-loss halt; live, each strategy has a 10% intraday daily-loss cap plus the account 2% kill switch, but the per-trade numbers here do not reflect that intra-day halt.

3.5 Regime filter

The strategy references SPY and requires *peer(SPY, vol_rank(20,60)) ≥ volrank_min*; with **volrank_min = 0.0** the gate is **OFF**. Our own evidence cuts against a naive high-vol prior: trades in the high-vol SPY bucket netted **-$68,254 (-10.4 bps/trade)**, while low-vol trades netted **+$68,844 (+7.6 bps/trade)**. The mechanical edge appeared in **low-vol** regimes here — the opposite of the Gao et al. high-vol result and a small-sample observation. Keep the gate off by default; any threshold is an experiment.

3.6 Transaction cost assumptions

Commission: $0 (Alpaca). Per-side friction: **half-spread 2 bps + slippage 3 bps = 5 bps per side**, i.e. **10 bps round-turn**. The dossier measures mega-cap half-spreads ~0.67-1.67 bps and off-radar mid-caps ~15.0 bps; stocks-in-play are liquid high-RVOL names, so 2 bps half-spread is a midpoint — but these names can widen on the catalyst move, hence stress:

Cost case	Round-turn	Net P&L; on basket
Base (1x)	10 bps	+$590
Stress (2x)	20 bps	-$155,439
Stress (3x)	30 bps	-$311,468

The strategy is extremely cost-sensitive: it survives only if realized round-turn stays near 10 bps; any systematic widening flips it negative.

4. Expected Performance

The table labels provenance. **“Authors' own”** = the paper's backtest, not replicated. **“Our run”** = this repo's mechanical basket test. **“Literature”** = a verified reference range, not a promise.

4.1 Headline numbers by provenance

Source	Variant	Key result	Provenance
Gao et al. 2018	SPY last-30-min timing	Sharpe 1.08 (vs B&H; 0.29)	Literature (peer-reviewed)
Aleti/Bollerslev/Siggaard 2025	SPY 15-min ML timing	After-cost Sharpe 1.37; FF6 alpha 20.83%	Literature (peer-reviewed)
Zarattini et al. 2024	QQQ-only ORB	Sharpe 1.12; win 24%; 0.13R/trade	Authors' own (working paper)
Zarattini et al. 2024	Top-20 stocks-in-play	Sharpe 2.81; >1,600% total	Authors' own (cost-light)
Our run	**Fixed 35-name basket, mechanical ORB**	**1,046 trades; win ~18%; net Sharpe 0.007; +$590**	**Our own backtest**

4.2 Our own backtest run

Setup: Alpaca v2 *iex* feed, 2024-01-01 to 2026-10-16, 700 trading days, 35 symbols, $100,000 account, 0.5% risk/trade, 0.10xATR14 stop, 10R placeholder, 15:50 flatten, 10 bps round-turn; regime and first-candle filters OFF.

Metric	Value
Trades	1,046 (524 long / 522 short)
Gross P&L;	+$156,619
Net P&L;	+$590
Gross annualized Sharpe	1.793
Net annualized Sharpe	0.007
Net t-stat	0.011
Max drawdown	$12,274
Bootstrap 95% CI (mean daily return)	[-0.00155, +0.00147]
Bootstrap 95% CI (Sharpe)	[-1.34, +1.10]

By year

Year	Trades	Win %	Net $	Exp bps	Avg RVOL
2024	339	20.1%	+16,557	+3.0	3.7
2025	388	14.2%	-48,144	-8.4	3.4
2026	319	20.1%	+32,177	+7.5	3.5

By side

Side	Trades	Win %	Net $	Exp bps
Long	524	17.6%	+22,047	+2.8
Short	522	18.2%	-21,457	-2.8

By SPY volatility regime

Regime	Trades	Win %	Net $	Exp bps
High-vol SPY	489	14.5%	-68,254	-10.4
Low-vol SPY	557	20.8%	+68,844	+7.6

By SPY daily direction

Day type	Trades	Win %	Net $	Exp bps
SPY up day	587	17.4%	-1,250	-0.1
SPY down day	459	18.5%	+1,840	+0.3

4.3 Realistic expectations for the deployed version

- **Trade frequency:** the basket generated ~1.5 trades/day (1,046 over 700 days) with a 3-trade cap. A news-selected long-only universe trades less — plausibly **0-2 qualifying trades per day, a handful per week**. The free IEX feed is sparse and may fail volume filters.
- **Expectancy:** the only cost-honest anchor is QQQ ORB: ~24% win, +0.13R/trade, Sharpe ~1.12. Our long side shows a low ~18% hit rate with positive skew. A realistic literature-based expectancy is **small per-trade expectancy, low win rate, positive skew** — not the 2.81 figure.
- **What we will not claim:** we do not expect 1,600% returns. Our bootstrap CI straddles zero; the news-selection edge is unvalidated until the dashboard backtest runs.

5. Validation Methodology

5.1 Data requirements

- **Intraday bars:** 1-minute consolidated/SIP bars; minimum **5-8 years** for a study of this kind (our run uses ~2.8 years, which is short). Use the same feed in backtest and live.
- **Daily history:** at least 60 prior sessions for ATR14/ADV14 and vol_rank(20,60); the loader caps lookback at 250 sessions.
- **Data fingerprints:** every run stores a hash of strategy YAML, risk snapshot, engine source, and input data for exact replay.

5.2 Walk-forward procedure

Yearly folds (train 2024, test 2025; roll forward). The base strategy has no fitted parameters beyond the paper's thresholds, so nothing is re-fit; experimental gates (volrank_min, candle_atr_min, rvol_min) must be selected on the training fold only and evaluated once on the holdout. A holdout used repeatedly counts as a research trial.

5.3 Correct cost and execution modeling

- **Next-bar fills:** an entry cannot fill on its decision bar; it fills on the next bar open plus bounded slippage.
- **Stop vs limit:** conservative stop-first brackets — on a bar touching both, the stop fills first.
- **Borrow:** short-side borrow fees/availability are not modeled in the standalone script — another reason to ship long only.
- **Costs:** charge half-spread, slippage, and commission on every fill; stress at 2x and 3x.

5.4 Pitfalls

- **Look-ahead:** never fill at the crossing-bar close; daily functions read completed sessions only; news gates use recorded signals at original timestamps.
- **Survivorship:** the 35-name basket is today's list with no delisted names; a point-in-time constituent list is a known gap.
- **Selection bias:** the whole edge depends on which names the news pipeline flags; validating it requires dashboard recorded-signal replay, not a static basket.
- **Overfitting:** tuning the experimental gates on the same 700 days is how spurious edges get invented. Keep all trials, including failures.

5.5 Statistical tests

Report t-stats with **Newey-West** adjustment or a **block bootstrap** of daily returns (1000x 10-day blocks here); **permutation tests** where applicable (Mesfin uses $p < 0.001$ as a gate); and the **Deflated Sharpe Ratio** (Bailey and López de Prado) to correct for the number of trials. Do not claim significance from a short sample or a Sharpe chosen after many experiments.

6. Infrastructure & Feature Gap Analysis

Nine required components are graded across four platforms as Provided / Partial / Gap.

Component	Thinkorswim	TradingView	Python	llamaOS
Real-time 1-min feed	Partial	Partial	Partial	Provided (Alpaca bars/news; IEX now, SIP path)
Opening-range calc	Partial	Partial	Partial	Provided (or_high/or_low(5) builtins)
Breakout order execution	Partial	Gap (alerts only)	Partial	Provided (marketable-limit brackets)
Regime filter (SPY vol_rank)	Gap	Partial	Gap	Partial (builtin; gate default OFF/unvalidated)
Sizing (0.5% risk)	Partial	Gap	Partial	Provided (risk-based shares, max_position cap)
Automated stop/target	Provided	Partial	Gap	Provided (broker-side brackets)
EOD flatten (15:50)	Partial	Gap	Gap	Provided (flatten_at 15:50; 15:55 risk, calendar-aware)
Trade logging / audit	Partial	Gap	Gap	Provided (hash-chained audit log)
Walk-forward backtesting	Gap	Gap	Partial	Partial (holdout, 2x/3x stress, bootstrap; formal fitting still a gap)

Already provided by llamaOS (per README): broker-side brackets; EOD flatten at 15:50 (15:55 risk deadline, early-close aware); the risk guard (Postgres limits outside model reach); the kill switch (daily-loss / reconciliation / stale-feed / manual; host-only reset); the hash-chained audit log; and the dashboard Backtesting workspace (holdout, 2x/3x cost stress, seeded bootstrap).

Remaining gaps: historical point-in-time universe constituents, row-level feed/adjustment provenance, quote/depth and partial-fill modeling, short borrow accounting, formal walk-forward parameter fitting, and a per-strategy daily-loss circuit breaker in the research model.

7. Improvement Roadmap

1. **Validate the news-selected universe first.** Run the news-gated strategy in the dashboard Backtesting workspace (holdout, 2x/3x stress, bootstrap) — the single most important upgrade, the untested premise.
2. **Entry-quality filters.** Breadth/VWAP/close filters (e.g. long requires price above opening VWAP) to skip weak breakouts; test on a separate fold.
3. **Execution optimization.** Use limit / stop-limit entries and avoid the opening auction; this attacks the 10 bps cost floor that determines profitability.
4. **Multi-sleeve uncorrelated instruments.** Add a second sleeve (e.g. Family B last-30-min SPY momentum) so daily P&L; is not carried by one regime.
5. **Regime-adaptive parameters.** Our low-vol observation suggests a vol_rank *down*-gate rather than the assumed high-vol gate — exploratory until confirmed out-of-sample.
6. **The 218-factor ML pipeline (Family E)** as a separate project: factor data layer, 15-min rebalancing, ensemble training. Do not fold into v1.
7. **Daily-loss circuit breaker.** Add an explicit per-strategy intraday daily-loss halt to the research model so backtest numbers match the live guard.

8. Deployment Guide (DEPLOY)

8.1 Where the strategy lives

- Long: *strategies/orb_stocks_in_play_pro_long.v1.yaml*
- Short mirror (gated): *strategies/orb_stocks_in_play_pro_short.v1.yaml*
- Rerunnable backtest: *scripts/research_orb_backtest.py*

8.2 Validate with the DSL loader/validator

From the repo root, using the factory risk file and the local test database:

```
RISK_FILE=config/risk.example.yaml \
DATABASE_URL=postgresql://llamaos:llamaos@127.0.0.1:5432/llamaos_test \
.venv/bin/python -m dsl.validator strategies/orb_stocks_in_play_pro_long.v1.yaml
```

Expected result (per reports/validation_output.txt): the **long** file validates ok=True (side long, window 09:35-10:30, flatten 15:50, risk 0.5%, references SPY); the **short** file validates ok=False with “entry.side 'short' is not allowed: allow_short is off on the Risk page.” That rejection is the intended default.

8.3 Backtest via the dashboard Backtesting workspace

- Open **Backtesting** and select the exact registered strategy version.
- Set inclusive ET session dates, starting USD equity, the symbol basket, and a benchmark.
- Set execution assumptions (half-spread, slippage, commission).
- **Check data coverage**; news gates use recorded signals at original timestamps.
- Run a **holdout window**, **2x and 3x cost stress**, and the **seeded bootstrap**.
- Review Saved runs; compare up to four configurations. Do not promote to paper until the news-selected universe clears the holdout.

8.4 Enabling the short side

Shorting is controlled by *allow_short* on the dashboard Risk page (factory default false, mirrored at Alpaca). Enabling it is a deliberate, documented experiment — our own evidence does not support it (short net -$21,457, -2.8 bps/trade).

8.5 Going to Alpaca paper

- Put paper credentials in *secrets/broker.env* against https://paper-api.alpaca.markets.
- Run *make init*, then *make paper-setup* (loads IEX history, validates/backtests, enables paper execution).
- Enroll the *paper-owner* account from *secrets/paper-owner.txt* in your authenticator.
- Run *make paper* and open the dashboard; set strategies to manual if you want per-trade approval.
- Keep MODE=record_only initially while the news pipeline builds point-in-time signal history.

Already provided: broker-side brackets, EOD flatten, the risk guard, the kill switch, and the hash-chained audit log. **Remaining gaps before live:** the dashboard validation of the news-selected universe (the standalone script only tested a fixed basket) and a per-strategy daily-loss breaker in the research model. Real-money execution is disabled

by design; paper-first is mandatory.

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- **No guaranteed profits.** Nothing promises, guarantees, or implies any specific return. Intraday trading is speculative and can lose some or all of invested capital.
- **Past and backtested performance.** Backtested results are hypothetical and have known limitations. Authors'-own figures cited from the literature (e.g. Sharpe 2.81, >1,600% returns) have not been independently replicated and used cost-light assumptions. Our own backtest nets approximately zero after costs with a confidence interval that includes negative Sharpe. Past performance does not guarantee future results.
- **Shorting, leverage, and market risk.** Short selling carries unlimited loss potential and borrow/recall risk. The news-selected universe concentrates in volatile, high-RVOL names that can gap sharply.
- **Data and execution risk.** Historical data may be incorrect, delayed, or revised; the free IEX feed differs from the consolidated tape. Live fills may differ materially due to slippage, partial fills, halts, depth, and network or broker outages.
- **Validation first.** The strategy's core premise — the news-selected universe — has not been validated out-of-sample. Do not trade real money until it passes the dashboard holdout and cost-stress tests and a sufficient paper-trading period.
- **Consult a licensed advisor.** Before committing capital, consult a properly licensed financial professional. Start in paper-trading mode. You are solely responsible for any trading decisions and their outcomes.